THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 12: **Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The angles of a quadrilateral are in the ratio $5 : 3 : 3 : 1$. The largest angle is: 1

(a) 160° **(b)** 140° **(c)** 170° **(d)** 150°

2. One angle of a parallelogram exceeds another adjacent angle by 70°. The smaller angle is: 1

(a) 55° **(b)** 90° **(c)** 65° **(d)** 50°

3. The diagonal of a rectangle is 34 cm and one side is 16 cm. Its area is: 1

(a) 510 cm^2 **(b)** 490 cm^2 **(c)** 480 cm^2 **(d)** 496 cm^2

4. The diagonals of a rhombus are 10 cm and 24 cm. Its perimeter is: 1

(a) 56 cm **(b)** 78 cm **(c)** 52 cm **(d)** 65 cm

5. The diagonals of a rhombus are 10 cm and 24 cm. Its area is: 1

(a) 130 cm^2 **(b)** 144 cm^2 **(c)** 120 cm^2 **(d)** 110 cm^2

6. The diagonals of a quadrilateral are 15 cm and 28 cm. The perimeter of the parallelogram formed by joining the midpoints of its sides is: 1

(a) 43 cm **(b)** 64 cm **(c)** 88 cm **(d)** 129 cm

7. In a parallelogram, two opposite angles are $(3x + 17)^\circ$ and $(x + 105)^\circ$. The value of x is: 1

(a) 45 **(b)** 44 **(c)** 46 **(d)** 43

8. In $\triangle ABC$, D and E are the midpoints of AB and AC and $DE = 30$ cm. Then BC is: 1

(a) 62 cm **(b)** 60 cm **(c)** 64 cm **(d)** 90 cm

9. The angles of a quadrilateral are in the ratio $5 : 2 : 6 : 2$. The largest angle is: 1

(a) 144° **(b)** 154° **(c)** 134° **(d)** 164°

10. One angle of a parallelogram exceeds another adjacent angle by 80°. The smaller angle is: 1

(a) 90° **(b)** 60° **(c)** 45° **(d)** 50°

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. One angle of a parallelogram is 20° more than its adjacent angle. Find all the angles. 2

12. A rectangle has a diagonal of 34 cm and one side 16 cm. Find its other side, area and perimeter. 2

13. The diagonals of a rhombus are 14 cm and 48 cm. Find its side, perimeter and area. 2

14. One angle of a parallelogram is 40° more than its adjacent angle. Find all the angles. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. The diagonal of a square is 18 cm. Find its area and its side in terms of $\sqrt{2}$. 3

16. In rhombus ABCD, $\angle A = 64°$. Find $\angle B$, and the angles $\angle OAB$ and $\angle OBA$ where O is the point of intersection of the diagonals. 3

17. In parallelogram ABCD, $AB = (2x + 7)$ cm, $CD = (x + 15)$ cm and $BC = (x + 6)$ cm. Find x , the sides and the perimeter. 3

OR

The diagonal of a square is 34 cm. Find its area and its side in terms of $\sqrt{2}$. 3

18. In rhombus ABCD, $\angle A = 98°$. Find $\angle B$, and the angles $\angle OAB$ and $\angle OBA$ where O is the point of intersection of the diagonals. 3

OR

In parallelogram ABCD, $AB = (2x + 1)$ cm, $CD = (x + 8)$ cm and $BC = (x + 2)$ cm. Find x, the sides and the perimeter. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. ABCD is a parallelogram with $AB = 11$ cm, $AD = 6$ cm and diagonal AC with $AC^2 = 157$ cm^2. Show that ABCD is a rectangle. 5

OR

The perimeter of $\triangle ABC$ is 82 cm. D, E, F are the midpoints of its sides. Find the perimeter of $\triangle DEF$ and the length of each of its sides in terms of the sides of $\triangle ABC$. 5

20. ABCD is a parallelogram with $AB = 13$ cm, $AD = 6$ cm and diagonal AC with $AC^2 = 205$ cm^2. Show that ABCD is a rectangle. 5

OR

The perimeter of $\triangle ABC$ is 56 cm. D, E, F are the midpoints of its sides. Find the perimeter of $\triangle DEF$ and the length of each of its sides in terms of the sides of $\triangle ABC$. 5

END OF QUESTION PAPER Total: 40 Marks

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